Aqua Edge: Simulation Progress Report

CSD457 Internet of Things | Breeti Bandyopadhyay | 22 October 2026

1. Scope

This report covers only the simulation part of Aqua Edge; the other parts of the project are reported separately by teammates. Simulation lets us create leaks, supply problems and contamination on purpose and see how a design reacts, which is not possible on a real building. Water behaviour is simulated in EPANET through WNTR (its Python library), the approach used by Pagano et al. [5] (EPANET and NS-3 co-simulation of a water network).

2. Reproducing the reference network

We reproduced the published SWI-FEED dataset of Pagano et al. [5] (EPANET and NS-3 co-simulation of a water network) exactly. Their NET_6 network (4,419 junctions, 5,066 pipes, 3 reservoirs) was run for 24 hours in WNTR and compared value by value with their published CSV files. This confirms our toolchain matches the reference paper, so later results rest on a trusted baseline.

Solver	Values compared	Largest difference
EPANET solver	105,816 pressures, 121,200 flows	0.37 m pressure
WNTR solver	105,816 pressures, 121,200 flows	0.000001 (CSV rounding only)

3. Leak experiments on the town network

On NET_6, neither leak rule from our references catches small leaks, and the two rules catch different large ones. We added leaks of 4 sizes at 8 random junctions and tested the pressure-drop rule of Alghamdi et al. [2] (pressure-drop leak rule and smart valve closure over LoRaWAN) and the supply-versus-branch mass balance of Ali et al. [6] (mass balance of supply vs branch flows for leaks, plus quality sensing), with realistic sensor errors.

Leak flow	Pressure-drop rule	Mass balance
0.2 L/s	1 of 8 (sensor noise, not the leak)	0 of 8
1.1 L/s	0 of 8	0 of 8
4.3 L/s	4 of 8 (1 a real drop)	2 of 8
10.5 L/s	3 of 8 (2 real drops)	5 of 8

The network is very stiff: even 65 to 77 L/s bursts near the pumps moved pressure by less than 5 psi. Only large bursts far from the main source create a visible low-pressure zone (Figure 1). The mass balance raised no false alarms but needs a leak of about 10% of a zone's inflow.

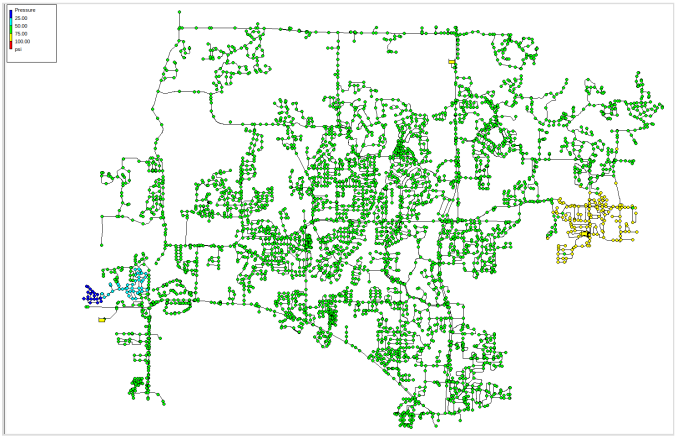


Figure 1. EPANET view of NET_6 with two main-pipe bursts on the far left. Pressure falls below 50 psi (cyan) and 25 psi (blue) only around the bursts.

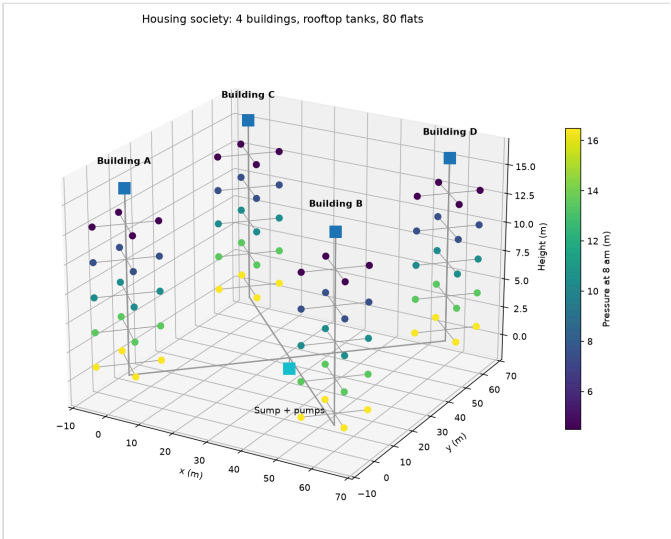


Figure 2. The housing-society network in 3D (4 buildings, rooftop tanks, 80 flats), flats coloured by pressure at the 8 am peak. Upper floors get less pressure.

4. Building-level network

Inside a building, a leak shows up in flow, not in pressure. Because a whole town is too coarse for one Aqua Edge node, we built our own housing-society network in EPANET: 4 buildings, ground plus 4 floors, 4 flats per floor (80 flats). Water goes from an underground sump, through one pump per building, to a rooftop tank, then down by gravity. Use is 540 L per flat per day (4 people at 135 L, the CPHEEO guideline) with morning and evening peaks.

Height is modelled through each junction's elevation, so pressure falls about 3 m per floor: 16.5 m on the ground floor and 4.5 m on the top floor at the morning peak (Figure 2). A 180 L/hour leak in one flat raised its building's night use (1 to 4 am) from 0.015 to 0.067 L/s, more than 4 times normal, but lowered that flat's pressure by only 0.22 m.

5. Trying out Webots

We installed Webots R2025a and built a first 3D model of one building to see whether it suits our node-level work. The model holds 20 flat nodes at their real floor heights, a tank node on the roof and a gateway on the ground (Figure 3). Each node is a Webots robot with a radio emitter and receiver, a status light, and its own Python program. Webots cannot simulate water flow, so it will not replace EPANET. Its value for us is running each node's own program in a 3D scene, with a radio link we can switch off to imitate an outage.

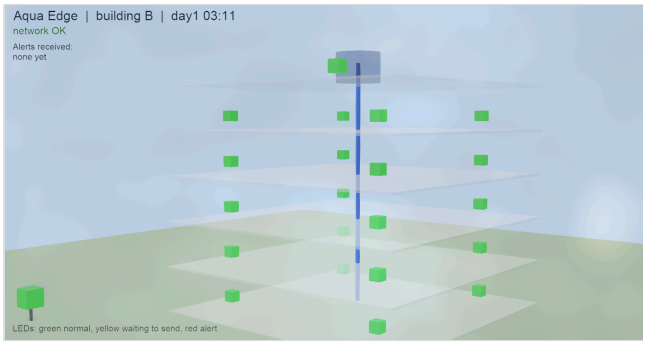


Figure 3. First Webots model of building B: 20 flat nodes (green boxes) on 5 floors, a tank node on the roof and a gateway on the ground (bottom left).

6. Plan

The next step joins the two tools: EPANET produces the water behaviour and Webots runs the node programs (Figure 4).

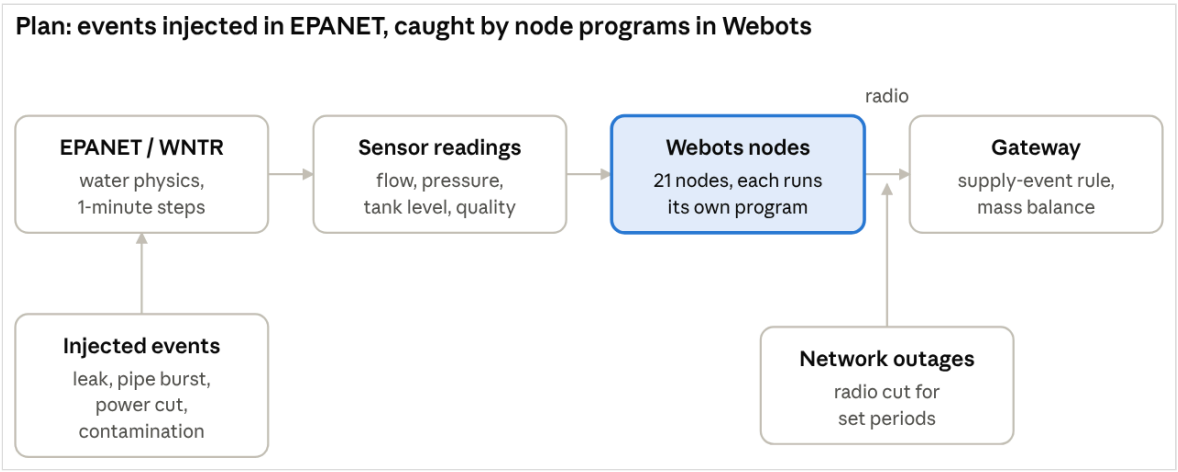


Figure 4. Planned simulation pipeline: events injected in EPANET, sensor readings passed to the Webots nodes, outages applied to the radio link.

1. Inject events in a 48-hour EPANET run of one building: day 1 normal, so nodes can learn their usual pattern; day 2 with a flat leak, a pipe burst and a power cut. Water quality readings are made up with contamination events added, since EPANET cannot model pH or TDS.
2. Turn the results into one sensor reading per node per minute and feed them to the Webots nodes.
3. Run the detection rules in each node: night flow and abnormal use, the mass balance of Ali et al. [6] (mass balance of supply vs branch flows for leaks, plus quality sensing), and a quality check standing in for Sharma et al. [1] (8-bit neural network on the device that classifies water quality), with the store-and-forward of Mariadhason et al. [4] (timestamped store-and-forward to SD card, duty cycling).
4. Cut the radio for set periods to imitate outages of growing length; later swap the simple Webots radio for NS-3 to model LoRa properly, as in Pagano et al. [5] (EPANET and NS-3 co-simulation of a water network).
5. Measure detection rate and delay, alert delivery against outage length, and false valve closures.